

PROJECT REPORT

LICENSED SOFTWARE REQUEST MANAGEMENT

ServiceNow Service Catalog and Workflow Automation

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Project Documentation

Prepared based on the implemented project and reference report structure.

Technology: ServiceNow

Application: LICENSED SOFTWARE REQUEST MANAGEMENT

Project: Automated Licensed Software Installation Request
Management

Table of Contents

1. INTRODUCTION

1.1 Project Overview

1.2 Purpose

2. IDEATION PHASE

2.1 Problem Statement

2.2 Empathy Map Canvas

2.3 Brainstorming & Idea Prioritization

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

3.2 Solution Requirements

3.3 Logical Data Flow

3.4 Technology Stack

4. PROJECT DESIGN

4.1 Problem–Solution Fit

4.2 Proposed Solution

4.3 Solution Architecture

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional Testing

6.2 User Acceptance Testing

6.3 Performance Testing

7. RESULTS

The following figures present the implementation evidence captured from the ServiceNow project. They cover catalog configuration, submitted variables, workflow automation, approvals, fulfillment, SLA/testing evidence, and deployment/update-set evidence.

7.1 Output Screenshots

7.2 Work Item Status Validation

8. ADVANTAGES & DISADVANTAGES

9. CONCLUSION

10. FUTURE SCOPE

11. APPENDIX

1. INTRODUCTION

1.1 Project Overview

The Licensed Software Installation Request system is a ServiceNow-based solution designed to simplify and automate the process of requesting installation of licensed software. Employees can submit software installation requests through a centralized Service Catalog. The system captures required information such as software name, software version, business justification, required date, employee, target device, manager, department, urgency, and license type.

After submission, ServiceNow creates the request records and processes the request through an automated workflow. The workflow retrieves catalog variables, evaluates the request, validates license information, routes the request for approval, and creates a Software Installation Catalog Task after the required approvals are completed. The task is assigned to the Software Support Team for fulfillment.

High-Level Flow

Employee → Service Catalog → REQ/RITM → License Validation → Approval → Software Asset Approval → Catalog Task → Software Support Team → Installation → Notification → Closure

Proposed Solution — Licensed Software Request Management

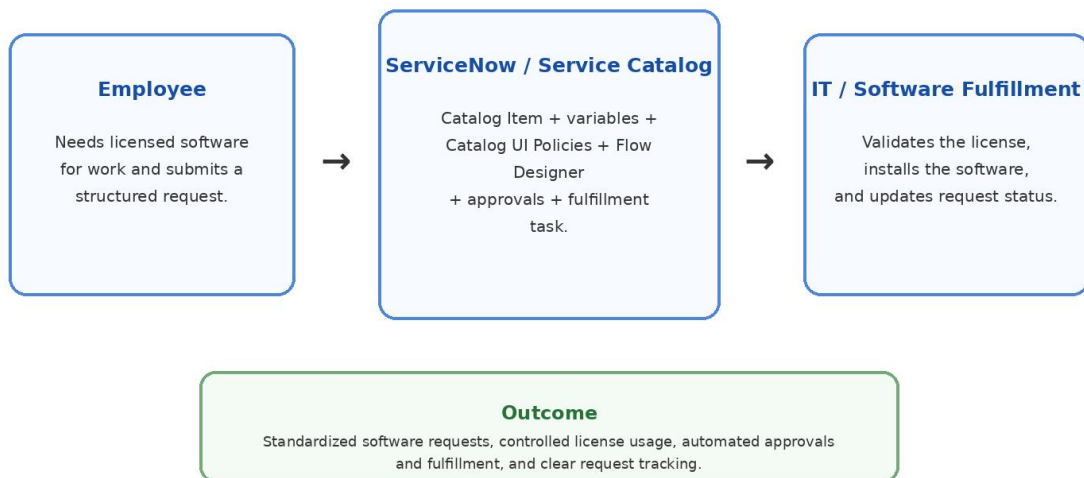


Figure 1: High-level workflow of the Licensed Software Installation Request System.

1.2 Purpose

The purpose of this project is to provide a centralized and structured process for managing licensed software installation requests.

The system aims to:

- Simplify licensed software request submission.
- Capture complete software and employee information.
- Validate license availability before fulfillment.
- Automate approval processing.
- Automatically create software installation tasks.
- Assign tasks to the Software Support Team.

- Improve request tracking and visibility.
- Provide SLA and notification support.
- Maintain controlled access through ServiceNow roles and ACLs.

2. IDEATION PHASE

2.1 Problem Statement

Employees may face difficulties when requesting licensed software because requests can be submitted through unclear or unstructured processes. Incomplete information, missing license checks, manual approvals, and manual task assignment can cause delays in software installation and make request tracking difficult.

Problem Statement Canvas

Parameter	Description
User	Employees requesting licensed software
Problem	Software requests may be incomplete or lack licensing information
Need	A simple and structured request process
Impact	Delays in validation, approval, and installation
How Might We?	How might we provide a centralized system for requesting, validating, approving, and installing licensed software efficiently?

2.2 Empathy Map Canvas

Employee / End User Perspective

Dimension	Employee Perspective
Says	I need a simple way to request the software I require.
Thinks	I want a clear form with relevant software and installation options.
Does	Opens the portal, selects the Licensed Software Installation Request, and submits details.
Feels	Wants fast, reliable, and transparent software support.
Pain Points	Incomplete forms, unclear approval status, license uncertainty, and delayed fulfillment.
Needs / Goals	Simple submission, clear tracking, timely approval, and successful installation.

2.3 Brainstorming & Idea Prioritization

Idea	Priority
Licensed Software Request Catalog Item	High
Catalog Variables	High
Catalog UI Policies	High
Catalog Client Scripts	High
License Validation	High
Approval Process	High
Flow Designer / Workflow Automation	High
Software Installation Task	High
Software Support Team Assignment	High

Notifications	Medium
SLA Monitoring	Medium
Reports and Dashboards	Medium
ACLs and User Roles	High

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The employee journey begins when the user accesses the Service Portal and ends when the licensed software request is fulfilled.

Stage	Employee Action	System Response
Access	Opens Service Portal	Portal displays available services
Discover	Finds Licensed Software Installation Request	Catalog item is displayed
Submit	Enters request details	Variables capture information
Validate	Completes required fields	UI Policies and Client Scripts validate the form
Process	Submits request	REQ and RITM are created
License Check	Waits for validation	Workflow checks configured license information
Approval	Waits for approval	Approval workflow is triggered
Fulfillment	Request is processed	Software Installation Task is created
Completion	Receives completed service	Task and request status are updated

3.2 Solution Requirements

Functional Requirements

ID	Requirement
FR-01	Employees shall access the Licensed Software Installation Request Catalog Item.
FR-02	The system shall capture required software and requester information.
FR-03	The system shall validate mandatory fields.
FR-04	Relevant fields shall be displayed or controlled dynamically.
FR-05	The system shall create REQ and RITM records after submission.
FR-06	The request shall undergo license validation.
FR-07	The request shall be sent for approval.
FR-08	A Software Installation Task shall be created after required approval.
FR-09	The Software Support Team shall process the assigned task.
FR-10	The system shall send relevant notifications and update request status.

Non-Functional Requirements

ID	Requirement
NFR-01	The system should provide a simple user interface.
NFR-02	The system should use ServiceNow access

	controls.
NFR-03	The workflow should process requests consistently.
NFR-04	The system should provide normal ServiceNow response performance.
NFR-05	The configuration should be maintainable and deployable through an Update Set.

3.3 Logical Data Flow

Employee Input → Catalog Variables → Service Catalog → REQ/RITM → Workflow → License Validation → Approval → Catalog Task → Software Support Team → Task Completion → Notifications / Dashboard

Proposed Solution — Licensed Software Request Management

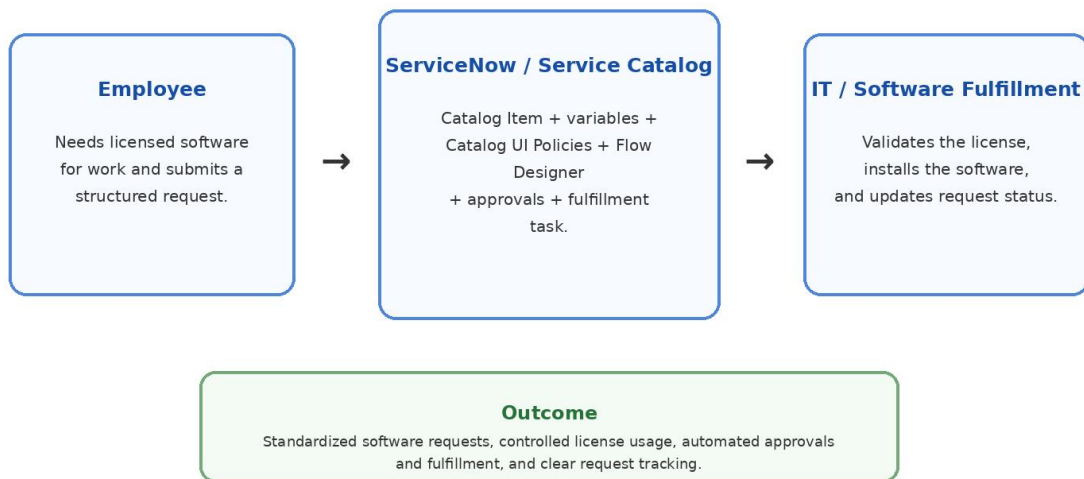


Figure 3: Logical data flow of the Licensed Software Installation Request process.

3.4 Technology Stack

S.No.	Component	Description	Technology
1	User Interface	Employee-facing request interface	Service Portal / Service Catalog
2	Catalog	Licensed software request submission	Service Catalog
3	Variables	Software and installation information	Catalog Variables
4	Rules	Dynamic fields and validation	Catalog UI Policies / Client Scripts
5	Automation	Approval and workflow processing	Flow Designer / Workflow
6	Data	Request and fulfillment records	REQ / RITM / Catalog Tasks
7	References	User and group information	sys_user / sys_user_group
8	Licensing	Software license validation	License / software asset records
9	Communication	Request event messages	ServiceNow Notifications
10	Monitoring	Time-based processing	SLA

		targets	
11	Security	Controlled access	Roles / ACLs
12	Deployment	Configuration packaging	Update Set

4. PROJECT DESIGN

4.1 Problem–Solution Fit

Parameter	Project Fit
Problem	Employees need a simple and structured way to request licensed software.
Solution	A ServiceNow-based Licensed Software Installation Request system.
Customer Benefit	Guided request entry, approval visibility, and predictable fulfillment.
Operational Benefit	Structured requests improve licensing checks, approvals, and installation.
Uniqueness	Combines Service Catalog, UI controls, license validation, approvals, automation, and task fulfillment.
Scalability	Additional software products, approval levels, and deployment integrations can be added later.

4.2 Proposed Solution

The proposed solution is a ServiceNow-based Licensed Software Installation Request system. Employees submit software installation requests through a dedicated Catalog Item. Catalog Variables capture request information, while UI Policies and Catalog Client Scripts support validation and dynamic form behavior.

After submission, the request is processed by SIRA_Software_Installation_Workflow. The workflow retrieves catalog variables, evaluates priority, performs license lookup, routes approval, and proceeds through software asset approval. Once the request is approved and validated, a Software Installation Catalog Task is created and assigned to the Software Support Team.

Notifications communicate approval, assignment, decision, and completion events. SLAs provide time-based monitoring, while the dashboard provides operational visibility.

4.3 Solution Architecture

The solution architecture connects the Service Portal, Licensed Software Request Catalog Item, catalog variables, validation rules, license information, approval process, workflow automation, Software Installation Tasks, notifications, SLAs, security, and dashboard reporting.

Service Portal → Catalog Item → Catalog Variables → REQ/RITM → SIRA Workflow → License Validation → Approval → Asset Approval → Software Installation Task → Software Support Team → Completion → Notifications / Dashboard

Solution Architecture — Licensed Software Request Management

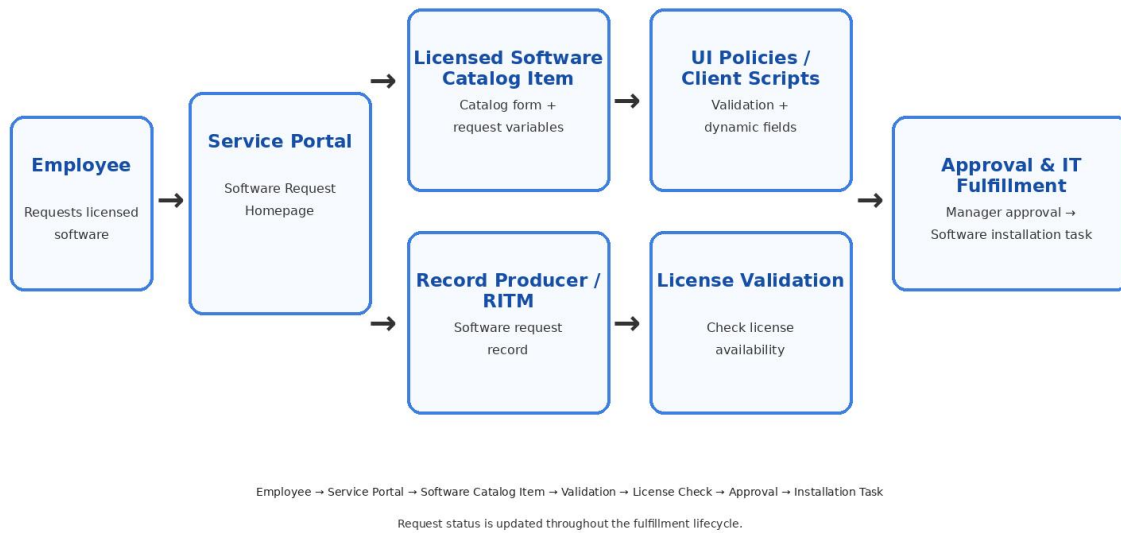


Figure 4: Solution architecture of the Licensed Software Installation Request System.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was divided into six implementation phases to provide a structured development and validation sequence.

Phase	Functional Area	Major Activities	Priority
Phase 1	Requirement Analysis & Planning	Requirements, roles, workflow, approvals, licensing, SLA and security.	High
Phase 2	Service Catalog & Form Configuration	Catalog item, variables, request records and form structure.	High
Phase 3	UI/UX Development & Customization	UI Policies, Catalog Client Scripts and form behavior.	High
Phase 4	Workflow Automation & Approval	Workflow, license validation, approvals and task creation.	High
Phase 5	Testing & Security	Functional testing, UAT, SLA, notification and security validation.	High
Phase 6	Deployment, Documentation & Final Presentation	Update Set, deployment validation, dashboard, documentation and final demonstration.	High

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional Testing

Test ID	Scenario	Expected Result	Status
TC-01	Open Service Portal	Portal loads successfully	Pass
TC-02	Find Licensed Software Request	Catalog item is visible	Pass
TC-03	Open request form	Form opens successfully	Pass
TC-04	Enter request details	Variables are available	Pass
TC-05	Validate fields	Mandatory validation works	Pass
TC-06	Submit request	REQ and RITM are created	Pass
TC-07	License validation	Configured license check is executed	Pass
TC-08	Approval process	Request goes for approval	Pass
TC-09	Create Software Installation Task	Task is created after approval	Pass
TC-10	Assignment	Task is assigned to Software Support Team	Pass
TC-11	Notification	Configured notification is generated	Pass
TC-12	End-to-end test	Complete workflow works	Pass

6.2 User Acceptance Testing (UAT)

UAT ID	Scenario	Acceptance Criteria	Result
UAT-01	Open Service Portal	Portal accessible	Accepted
UAT-02	Find Licensed Software Request	Catalog item visible	Accepted
UAT-03	Open form	Request form opens	Accepted
UAT-04	Enter software details	Required options available	Accepted
UAT-05	Validate fields	Required validation works	Accepted
UAT-06	Submit request	Request created	Accepted
UAT-07	License validation	License check works	Accepted
UAT-08	Approval	Approval works	Accepted
UAT-09	Task creation	Task assigned	Accepted
UAT-10	Complete request	Request completed	Accepted

UAT Conclusion

The tested Licensed Software Installation Request workflow satisfies the documented acceptance criteria for the implemented scope.

6.3 Performance Testing

The implemented workflow was tested for normal interactive usage within the ServiceNow development environment.

- Catalog form loading.

- Request submission.
- Catalog variable validation.
- Workflow execution.
- License validation processing.
- Approval workflow execution.
- Software Installation Task creation.
- Notification generation.
- Request completion.

Note: Large-scale concurrent-user load testing was outside the scope of the project.

7. RESULTS

7.1 Output Screenshots

Insert the actual project screenshots in the following locations, matching each caption.

The screenshot shows the ServiceNow interface for creating a 'Catalog Item - Software Installation Request'. The browser address bar shows the URL: `dev270509.service-now.com/now/nav/ui/classic/params/target/sc_cat_item.do%3Fsys_id%3D4130260847530310d799bb30116d4337%26sysparm_record_target%3Dsc_cat_item%26sysparm_record_row...`. The page title is 'Catalog Item - Software Installation Request'. Below the title, there are buttons for 'Copy', 'Try It', 'Update', 'Edit in Catalog Builder', and 'Delete'. A blue informational box states: 'Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies. Enter a Name and Short description to display for the item. Enter a Price, approvals, variables, and other information as needed.' The form fields are as follows:

Name	Software Installation Request	Application	Global
Catalogs	Service Catalog	Active	☒
Category	Software	Fulfillment automation level	Unspecified
State	Published		
Checked out	false		
Owner	System Administrator		
Secondary owners			

Below the form fields, there are tabs for 'Item Details', 'Process Engine', 'Picture', 'Pricing', and 'Portal Settings'. The 'Item Details' tab is selected, showing a 'Short description' field with the text 'Request to install company-approved software' and a 'Description' field with a rich text editor toolbar.

Figure 7.1:Licensed Software Installation Request - Catalog Item Configuration

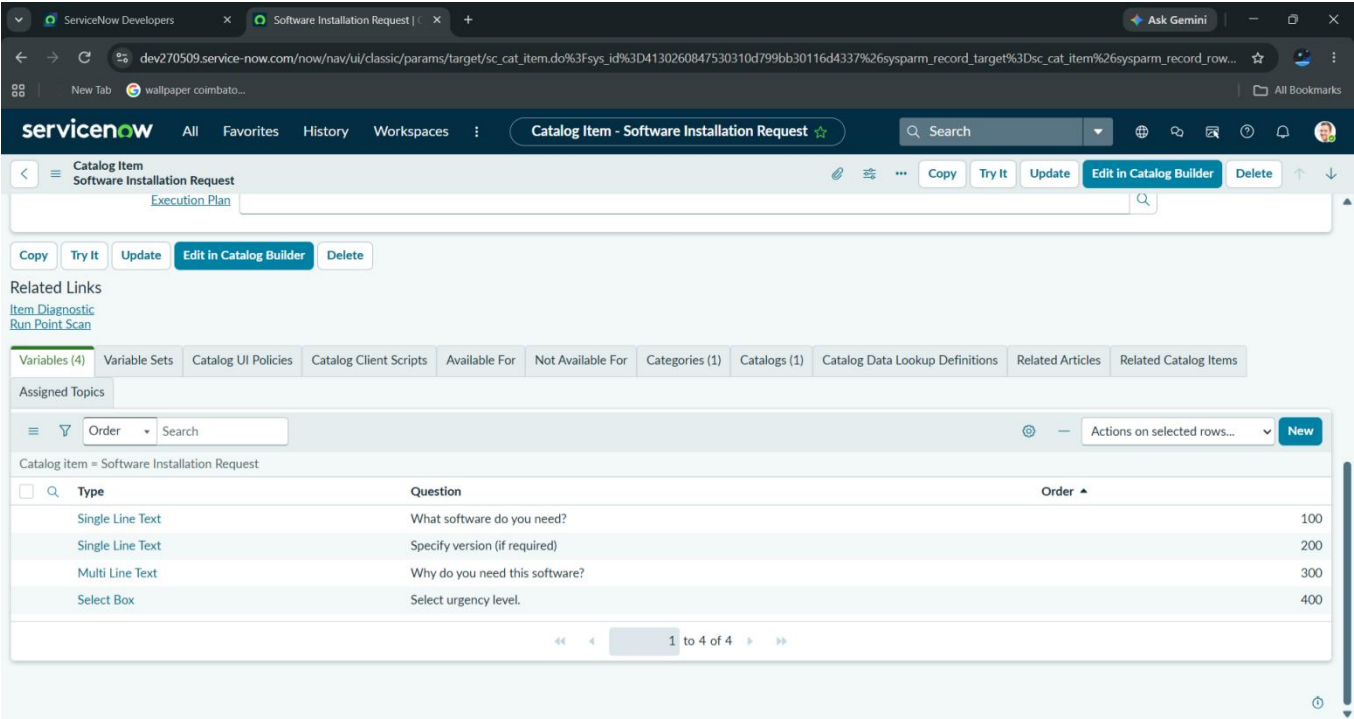


Figure 7.2: Licensed Software Installation Request - Configured Catalog Variables

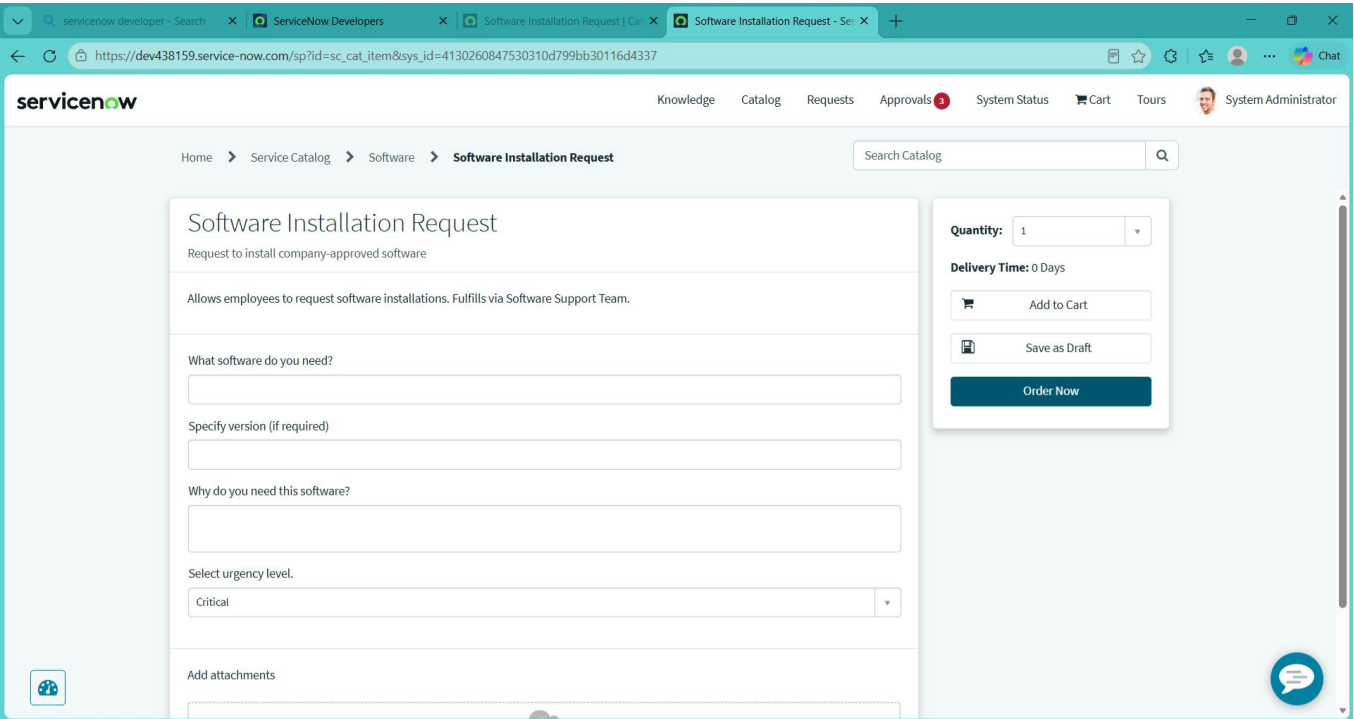


Figure 7.3: Licensed Software Installation Request - Service Portal Request Form

Software Installation Request
Request to install company-approved software

Allows employees to request software installations. Fulfills via Software Support Team.

What software do you need?
Microsoft Office 365

Specify version (if required)
2021

Why do you need this software?
Required for daily work

Select urgency level.
High

Quantity: 1
Delivery Time: 2 Days
Add to Cart
Save as Draft
Order Now

Figure 7.4: Licensed Software Installation Request - filled Service Portal Request Form

Request Summary - REQ0010006

Submitted : 2026-09-10 10:17:49
Request Number : **REQ0010006**
Estimated Delivery : 2026-09-10

Item	Delivery Date	Stage	Price (each)	Quantity	Total
Software Installation Request	2026-09-10		---	1	---

Total: \$0.00

Figure 7.5: Licensed Software Installation Request - Generated REQ/RITM Record

Requested Item
RITM0010006

Number: RITM0010006

Item: Software Installation Request

Request: REQ0010006

Requested for: System Administrator

Due date: 2026-09-10 10:17:49

Configuration item:

Watch list:

Opened: 2026-09-10 10:17:49

Opened by: System Administrator

Stage: Request Approved | request_approved

State: Open | 1

Quantity: 1

Estimated delivery:

Backordered:

Order Guide:

Variables

What software do you need? (Software Name)
Microsoft Office 365

Specify version (if required) (Version Required)
2021

Why do you need this software? (License Justification)
Required for daily work

Figure 7.6: Licensed Software Installation Request - Generated REQ/RITM Record

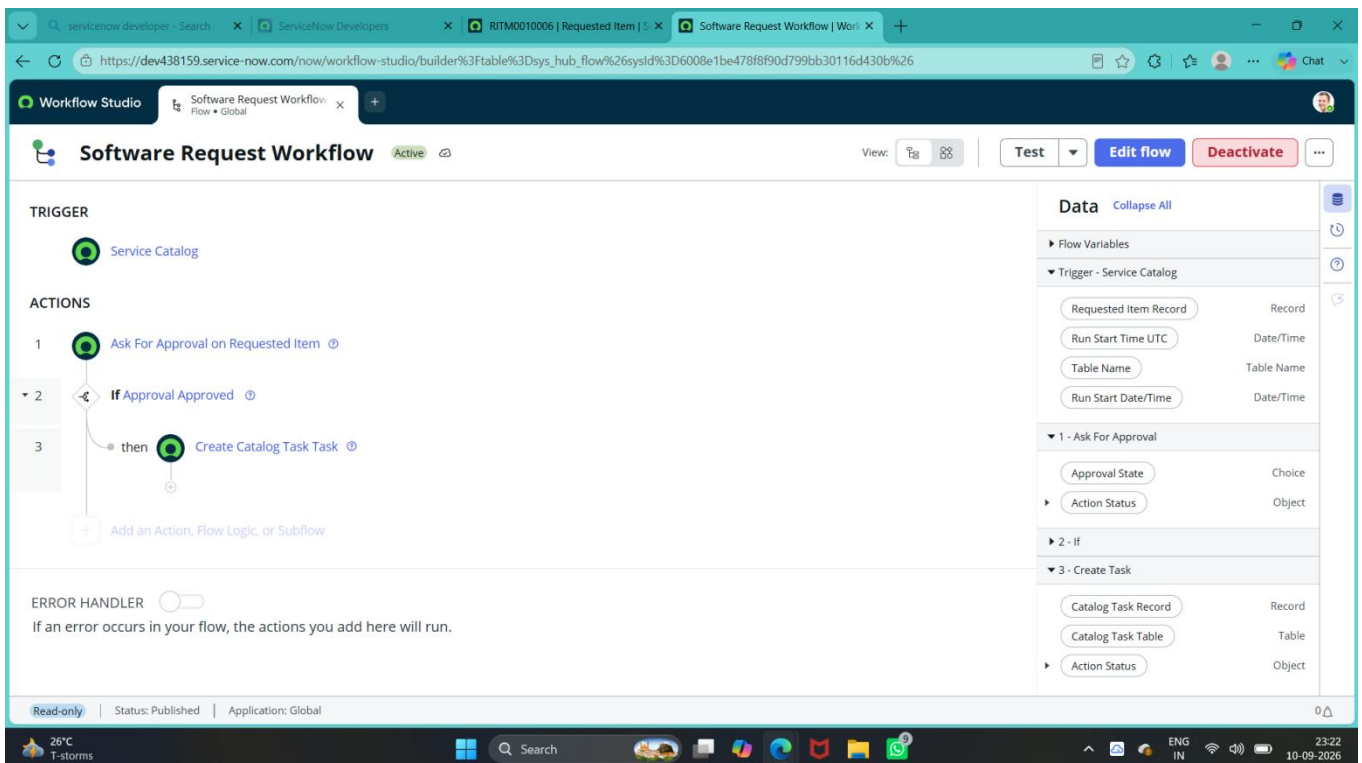


Figure 7.7: Software Request Workflow - Workflow Automation Configuration

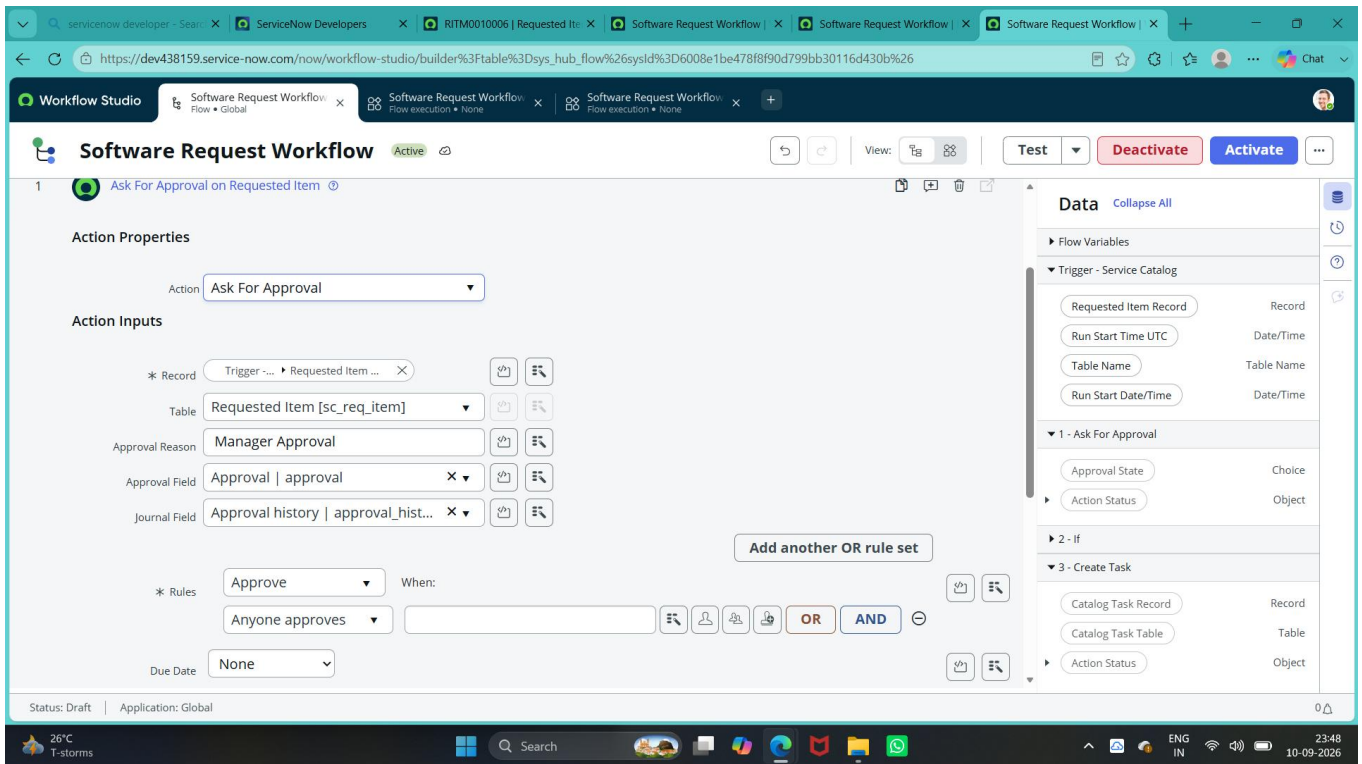


Figure 7.8: Software Request Workflow - Approval Configuration

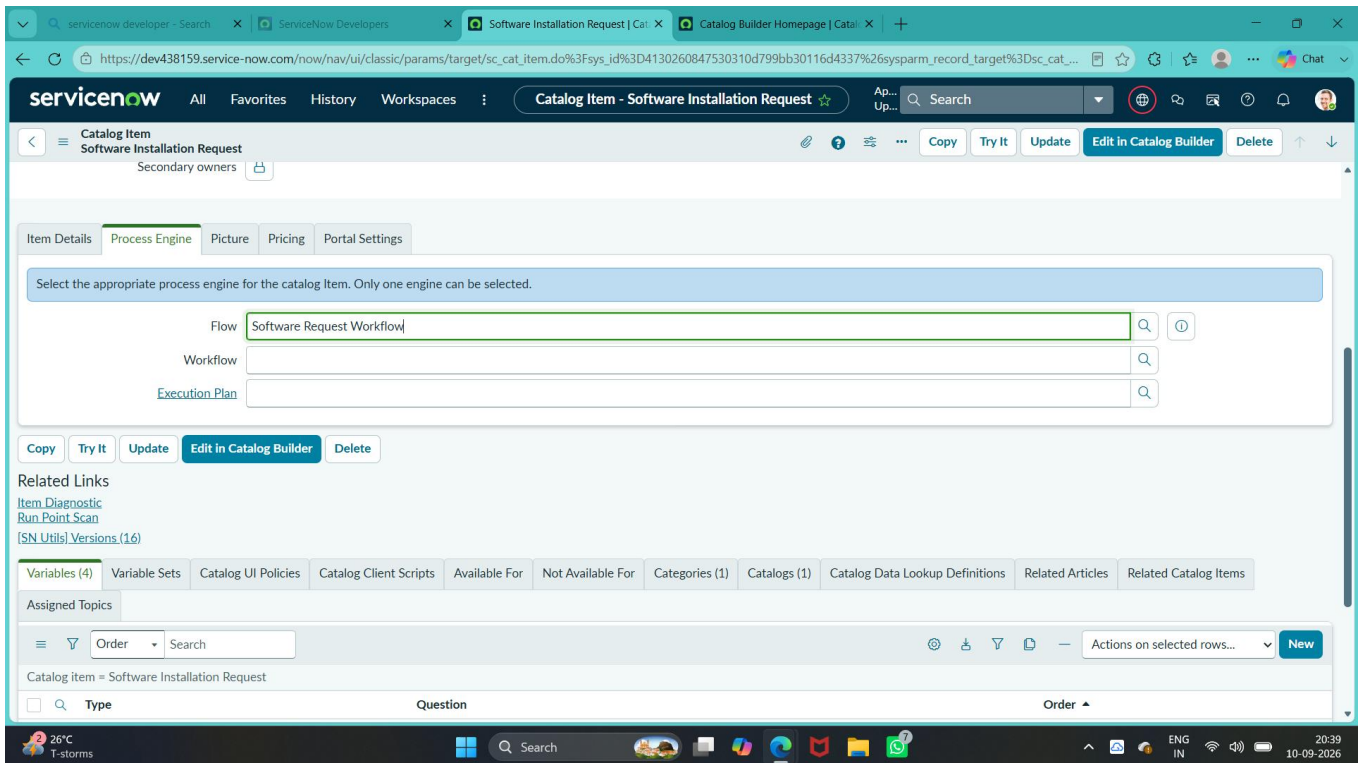


Figure 7.9: Software Installation Request - Process Engine Configuration

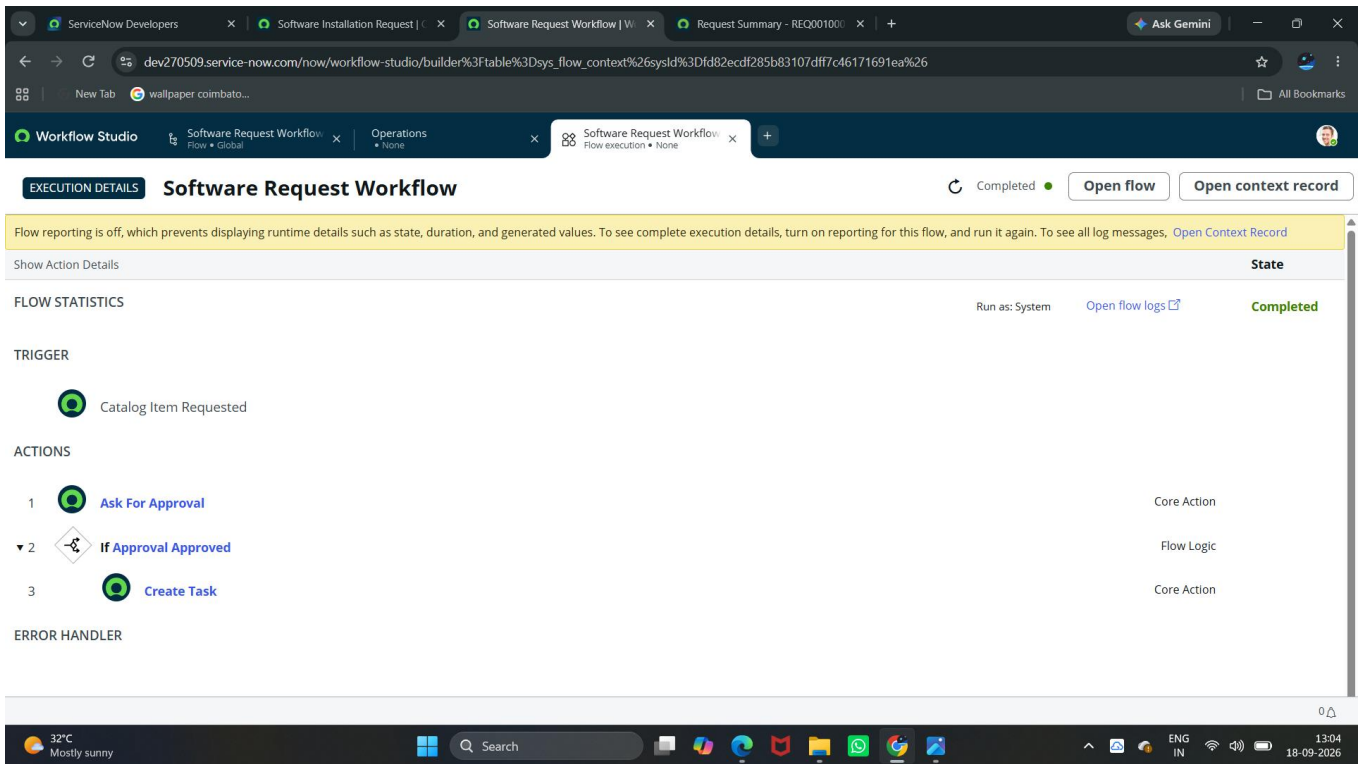


Figure 7.10: Software Request Workflow - Successful Flow Execution

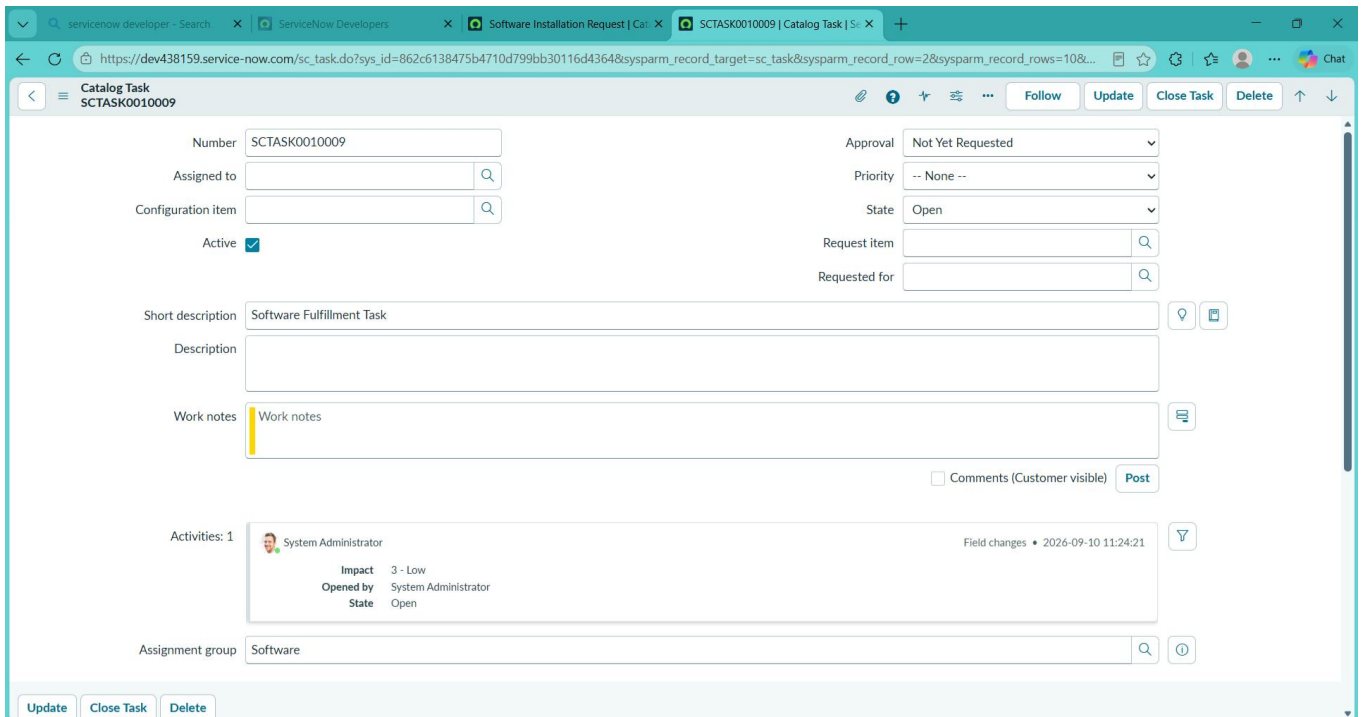


Figure 7.11: Software Fulfillment Task - Catalog Task Configuration

Name	Application	State	Installed from	Created	Created by	Parent	Batch Base
Default	Global	In progress		2026-04-30 00:40:06	system	(empty)	(empty)
Licensed Software Installation - Capstone	Global	Complete		2026-09-02 01:54:28	admin	(empty)	(empty)

Related Links
[Merge Update Sets](#)

Figure 7.12: Licensed Software Installation - Update Set Completed

Retrieved Update Set
 Licensed Software Installation - Capstone

All problems have been addressed. Click Commit Update Set below to continue.

After committing this update set, ensure you map any unmapped custom tables to a subscription so that you can track your entitlements accurately in Subscription Management.

Name	Licensed Software Installation - Capstone	Committed	
Application	Global	Inserted	50
Update source		Updated	2
Parent		Deleted	0
State	Previewed	Collisions	1
Loaded	2026-09-17 11:24:06	Total	53

Description: Update Set for Employee Request Installation of Licensed Software through Service Catalog capstone project.

Application name: Global

Update Delete Run Preview Again Commit Update Set

Related Links
[Show All Preview Records](#)

Figure 7.13: Licensed Software Installation - Update Set Preview Resolved

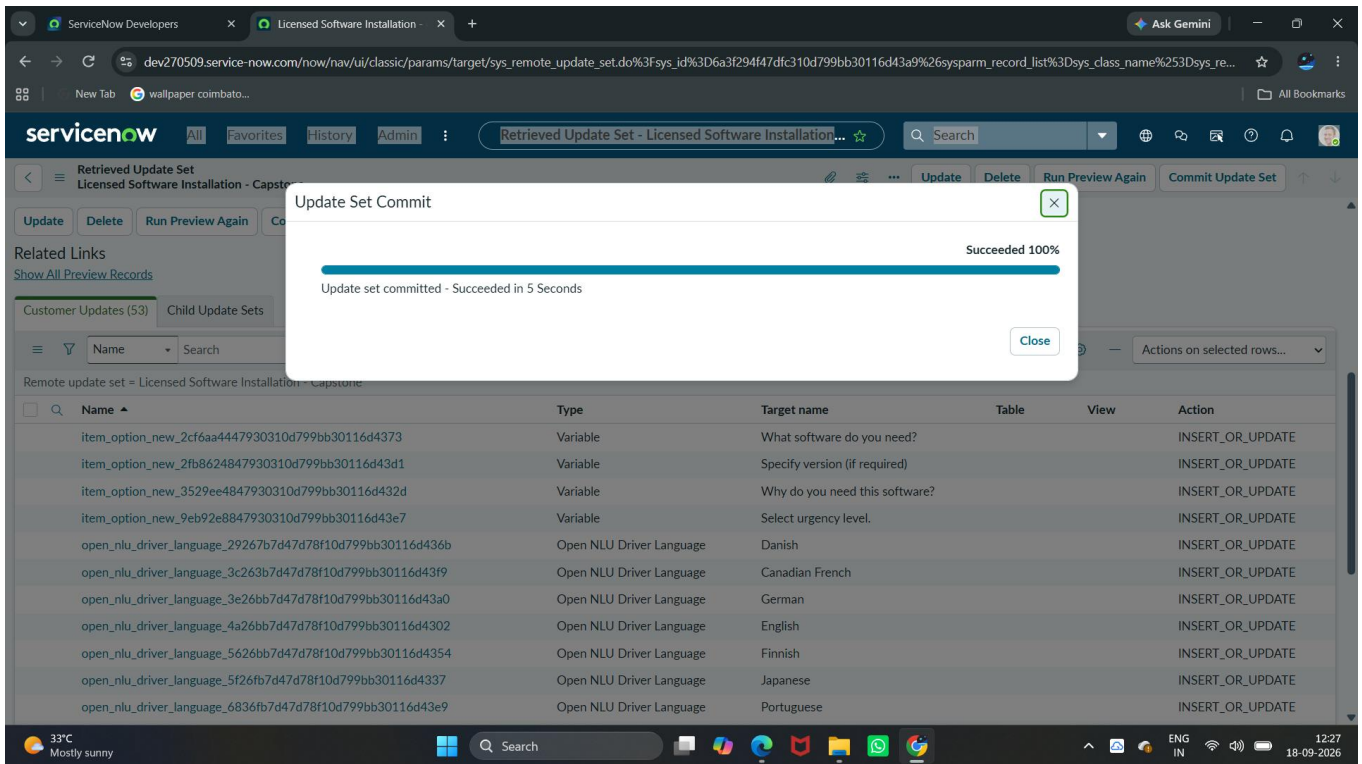


Figure 7.14: Licensed Software Installation - Update Set Committed

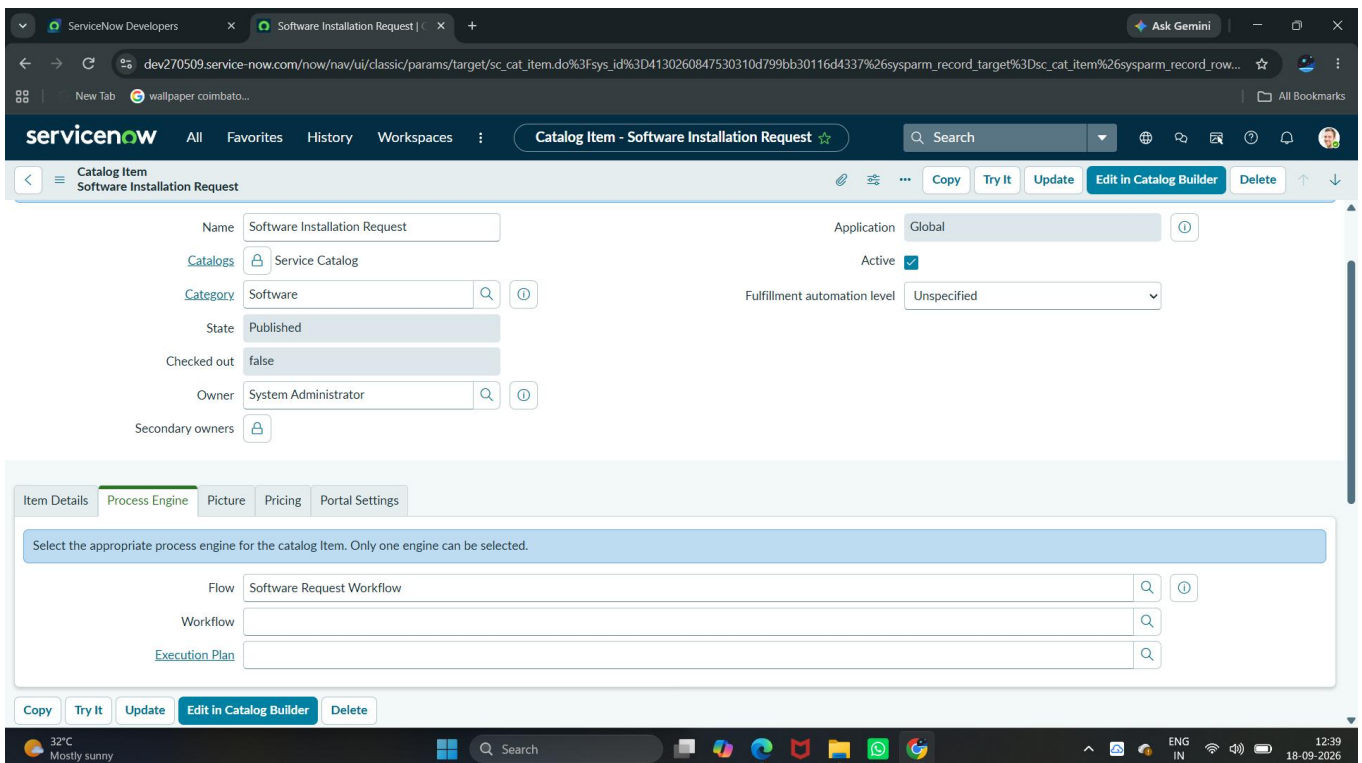


Figure 7.15: Target Instance - Software Installation Request Catalog Item

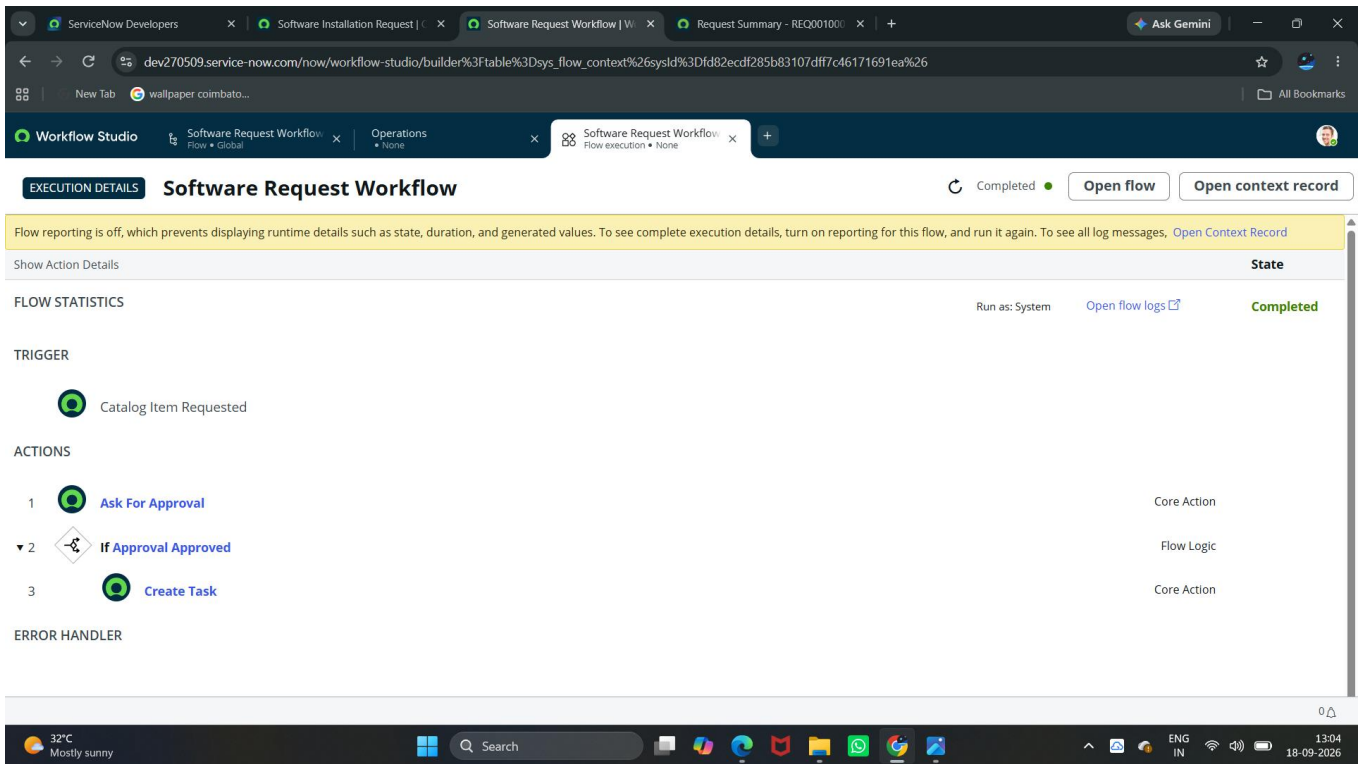


Figure 7.16: Target Instance - Software Request Workflow Execution

7.2 Work Item Status Validation

Component	Status	Validation
Licensed Software Request Catalog	Completed	Catalog item created and accessible
Catalog Variables	Completed	Required fields configured
Catalog Client Scripts	Completed	Validation configured
License Validation	Completed	License check configured
Approval Process	Completed	Approval configured
Software Installation Task	Completed	Task created for fulfillment
Software Support Team	Completed	Fulfillment assignment configured
Notifications	Completed	Request events communicated
Service Portal	Completed	Catalog accessible
Security	Completed	Roles and ACL controls configured
Update Set	Completed	Deployment package prepared

8. ADVANTAGES & DISADVANTAGES

Advantages

- Provides centralized licensed software request submission.
- Reduces incomplete request information.
- Improves request standardization.
- Automates license validation and approvals.

- Automatically creates fulfillment tasks.
- Improves visibility into the request lifecycle.
- Reduces manual effort for the Software Support Group.
- Supports controlled deployment and auditability.

Disadvantages

- Depends on correct ServiceNow configuration.
- Initial configuration requires ServiceNow platform knowledge.
- Complex workflows may require additional maintenance.
- License data must be accurate for reliable validation.
- Advanced deployment integrations may require additional development.

9. CONCLUSION

The Licensed Software Installation Request through the Service Catalog system provides a structured approach to managing licensed software installation requests using ServiceNow.

The solution enables employees to submit requests through a centralized catalog, captures required information, validates request and licensing data, supports approval processing, and automates Software Installation Task creation. Tasks are assigned to the Software Support Team for fulfillment, while notifications, SLAs, dashboards, roles, and ACLs improve operational visibility and control.

The project demonstrates how ServiceNow capabilities such as Service Catalog, Catalog Variables, UI Policies, Catalog Client Scripts, Flow Designer / Workflow, Approval Engine, license validation, Task Management, Notifications, SLA, Security, and Update Sets can be combined to improve operational efficiency and the employee experience.

10. FUTURE SCOPE

The system can be enhanced in the future with:

- Integration with automated software deployment tools.
- Advanced dashboards and reporting.
- Real-time license inventory synchronization.
- Automatic priority calculation.
- Additional approval levels for restricted or high-cost software.
- Automatic license renewal and reclamation.
- Email and mobile notifications.
- AI-based request classification.
- Virtual Agent support.
- Knowledge articles for common software requests.
- Integration with endpoint management platforms.

11. APPENDIX

Appendix A – Technology Components

- ServiceNow Service Catalog
- Catalog Items

- Catalog Variables
- Flow Designer / Workflow
- Approval Engine
- REQ and RITM Records
- License / Software Asset Information
- Software Installation Catalog Tasks
- sys_user
- sys_user_group
- Notifications
- Update Set

Appendix B – Project Flow

Employee Request → Service Catalog → Catalog Variables → REQ/RITM → License Validation → Approval → Software Asset Approval → Software Installation Task → Software Support Team → Installation → Notification → Closure